A coset geometry with a type cycle but no triality

Human Compression project: bounded recovery of L122

Abstract

We give an explicit rank-three coset geometry whose defining subgroups are cyclically permuted by a group automorphism, although no correlation of order three cycles its types. This refutes the bare existential assertion of Lemma 1 of Leemans–Stokes in its printed generality. The proof uses a small integer certificate, including a full correlation-group bound by color refinement; no graph-automorphism search is a proof premise.

1 The exact assertion and the construction

The fixed six-page source [1], physical pages 2–3, calls a rank-three pregeometry a geometry when every flag extends to a chamber. It defines a triality to have order three. Lemma 1 assumes only that an automorphism of the coset group cycles three defining subgroups; it does not require firmness, residual connectedness, or order three for that automorphism. We retain exactly this meaning. All group actions and permutation products below are on the right.

Let T be the subgroup of S_{18} generated by

$$u = (3\ 17\ 15\ 13\ 11\ 9\ 7\ 5)(4\ 18\ 16\ 14\ 12\ 10\ 8\ 6),$$
$$v = (1\ 2\ 11)(3\ 13\ 9)(4\ 15\ 6)(5\ 8\ 12)(7\ 18\ 16)(10\ 14\ 17).$$

The following two elements belong to T:

$$a = (1\ 2\ 3\ 5\ 16\ 15\ 14\ 9\ 11)(4\ 7\ 10\ 6\ 12\ 13\ 17\ 18\ 8),$$
$$b = (2\ 13)(3\ 14)(4\ 8)(5\ 11)(6\ 9)(7\ 16)(10\ 12)(17\ 18).$$

The certificate reconstructs T by closure under these literal generators: $|T| = 2448$, $|a| = 9$, $|b| = 2$, and $|\langle a^3, b\rangle| = 12$. The discovery model was $\mathrm{PSL}_2(17)$, but that abstract identification is not needed anywhere in the proof.

Put $G = \langle c\rangle \times T$, where $|c| = 3$, and set

$$H = \langle ca^3, b\rangle, \qquad H_i = H^{a^i} \quad (0 \le i < 3), \qquad \sigma(g) = a^{-1}ga.$$

Here T denotes its embedded factor in G. Direct closure gives

$$|G| = 7344, \quad |H_i| = 12, \quad H_i \cap H_j = B = \langle ca^3\rangle\ (i \ne j), \quad \langle H_0, H_1, H_2\rangle = G. \tag{1}$$

It also verifies that the three H_i are distinct and $H^{a^3} = H$. Consequently σ cyclically permutes them, as required in the source.

Let Γ have the 612 right cosets of H_i as its objects of type i. Cosets of different types are incident when they intersect. This is indeed a geometry: an incident pair has a common element g and extends to (H_0g, H_1g, H_2g); every singleton extends in the same way, and a three-element flag already is a chamber. Its incidence graph is connected by (1). The certificate constructs all coset partitions, hence every incidence, rather than sampling them. There are 1836 objects, 7344 edges, valency eight, and 2448 chambers. Each panel has one completion. Thus this example is not firm or residually connected; neither property is a hypothesis of the printed lemma. In fact the right action of G is chamber-transitive, since its chamber orbit has size $|G : B| = 2448$.

2 A central order-nine type cycle

Write R_g for right multiplication by g on all cosets. This action is faithful. Indeed its kernel lies in B, and the certificate checks that neither of the two nonidentity elements of B fixes all cosets. The induced correlation

$$\tau(H_i g) = H_{i+1} a^{-1} g a$$

cycles the types. Define $z = \tau R_{a^{-1}}$, so

$$z(H_i g) = H_{i+1} a^{-1} g. \tag{2}$$

It commutes with every R_g. Since $ca^3 \in H_i$ for every i,

$$z^3(H_i g) = H_i a^{-3} g = H_i c g = H_i g c; \qquad z^3 = R_c. \tag{3}$$

Thus z has order nine. Further, $\langle z \rangle \cap R_T = 1$: an element of this intersection fixes types, so it is $z^{3j} = R_{c^j}$, and faithfulness of $\langle c \rangle \times T$ gives $j = 0$ modulo three. Therefore the correlation group contains

$$L = \langle z, R_T \rangle \cong C_9 \times T, \qquad |L| = 22032. \tag{4}$$

Every element of order three in L preserves all types. To see this, write it uniquely as $z^e R_t$. Its cube being the identity forces $z^{3e} = R_{t^{-3}}$; the trivial intersection just proved gives $3 \mid e$, exactly the condition for preserving types.

3 The full correlation bound

Let A be the full correlation group: incidence-preserving permutations which send each type class to a type class. Form an auxiliary graph by adding three marker vertices m_0, m_1, m_2, with m_i adjacent to every object of type i. Color the objects with one color and all three markers with a second color. The markers are *not* individually colored. Automorphisms of this colored graph are exactly A: the unique marker neighbor of an object records its type, and any permutation of the three markers is allowed. Thus this construction imposes no extra order or type-fixing requirement on a correlation.

Lemma 1. *Refine a vertex coloring by replacing each color with the pair consisting of its previous color and its vector of neighbor counts in the previous color classes. Every color-preserving automorphism preserves every refined class. If refinement after individualizing vertices is discrete, their pointwise stabilizer is trivial.*

Proof. An automorphism preserves a vertex's previous color and bijects its neighbors in every previous color class. It therefore preserves the refined color. Induction proves the first assertion; singleton classes prove the second. □

For reproducibility the certificate numbers objects from zero. Group elements are sorted lexicographically as (j, t) with $0 \le j < 3$ and t its 18-entry permutation tuple; right cosets are numbered by the first uncovered representative, separately for each type. The markers are 1836, 1837 and 1838. Vertices 0 and 612 are incident. After individualizing these two objects, the numbers of color classes at successive refinements are

$$9, 21, 60, 187, 579, 630, 633, 633.$$

The class containing vertex 1 is exactly $\{1, 441, 481\}$. After individualizing 0, 612 and 1, the sequence is

$$11, 28, 126, 786, 1822, 1839, 1839;$$

in particular it is discrete. All neighbor counts are recomputed from the complete coset incidence relation.

There are at most 1836 images of vertex 0. Once it is fixed, vertex 612 has at most eight images, its object neighbors. Once both are fixed, vertex 1 has at most three images by Lemma 1, and their pointwise stabilizer is trivial. Hence

$$|A| \le 1836 \cdot 8 \cdot 3 = 44064. \tag{5}$$

The data file additionally supplies one involution r, as a complete permutation of the 1839 vertices. The certificate checks every adjacency and both colors directly. On markers it fixes m_0 and interchanges m_1, m_2. Thus $r \notin L$, since elements of L induce only cyclic type permutations. Equations (4)–(5) prove $|A| = 44064$ and $[A : L] = 2$. They also show that the full type-preserving group is exactly R_G: the type image is all S_3, and its kernel has order $44064/6 = 7344$.

Theorem 2. *The geometry Γ satisfies the hypotheses of source Lemma 1, but has no correlation of order three cycling its types.*

Proof. The construction and (1) verify the exact source hypotheses. Since L has index two in A, every element of order three of A lies in L. The observation following (4) says that every such element fixes types. This contradicts the defining property of a triality. □

There is also a complete Sylow certificate, independent of a conjugacy class routine. The subgroup $P = \langle z, R_a \rangle$ is $C_9 \times C_9$ of order 81, the full 3-part of 44064. Its 81 explicitly generated permutations have order distribution $1 : 1$, $3 : 8$, $9 : 72$. All eight order-three elements preserve types. By Sylow conjugacy, the same holds for every order-three element of A, since the type-preserving kernel is normal.

4 Finite boundary and consequences

The whole `certificate.py` and `data.json` are required finite premises. They use only exact integers and Python's standard library. The former constructs the finite groups and all cosets, verifies faithfulness and every adjacency, computes both refinement sequences, checks the complete supplied reflection and enumerates the explicit 81-element Sylow subgroup. GAP and nauty were used to discover the model and reflection, but neither is called by this certificate. The search over other groups is not an exhaustiveness premise for this one example.

This resolves the previously isolated *bare existential* obligation of L122, not merely the order of the induced map. The valid repair of the source construction is its already proved order qualification: requiring σ^3 to act trivially on every defining coset, in particular $\sigma^3 = 1$, supplies an actual triality. The six explicit source-family examples already satisfy that stronger hypothesis and are unaffected. No whole-original admission, new conjecture answer, formalization or measurement is asserted by this bounded author packet.

References

[1] D. Leemans and K. Stokes, *Coset geometries with trialities and their reduced incidence graphs*, Acta Math. Univ. Comenianae **88**(3) (2019), 911–916.